

1 Use Case Diagram ★ ★ ★ (MOST IMPORTANT)

◆ What is it?

A Use Case Diagram shows **what the system does** from the **user's point of view**.

◆ Why do we use it?

- To understand **system requirements**
- To show **interaction between user and system**

◆ Example:

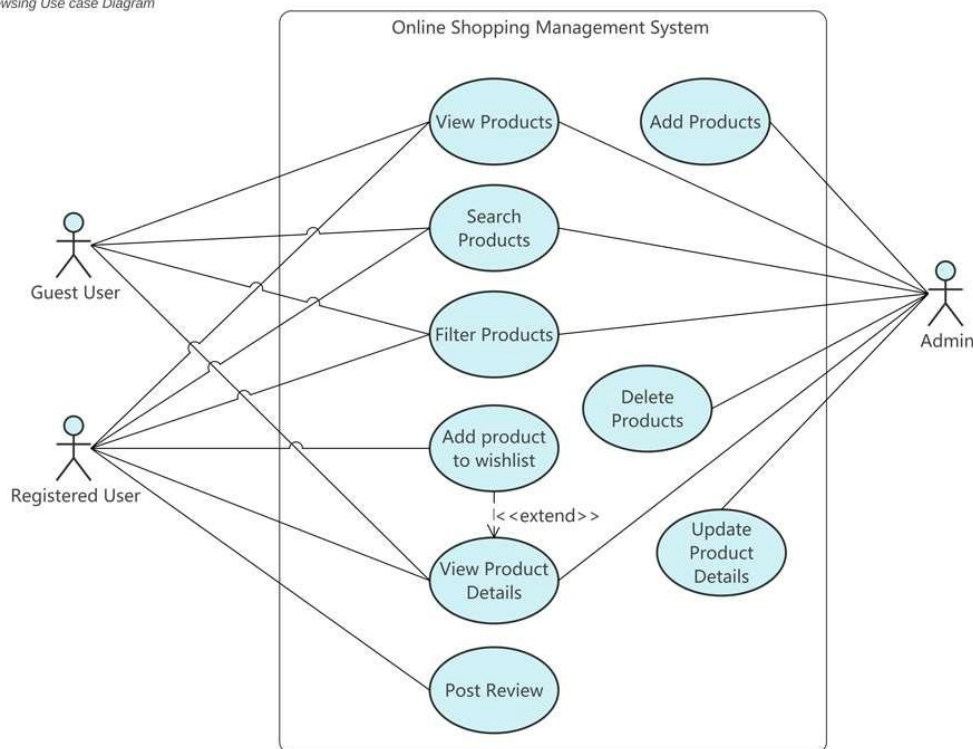
Student → Login, View Result

Admin → Add Student, Generate Report

◆ Viva one-liner:

“Use Case Diagram is used to represent system functionality and interaction between users and the system.”

Product browsing Use case Diagram



2 Class Diagram ★ ★ ★ (VERY IMPORTANT)

◆ What is it?

A **Class Diagram** shows **classes, attributes, methods, and relationships**.

◆ Why do we use it?

- For **system design**
- Acts as a **blueprint for coding**

◆ Example:

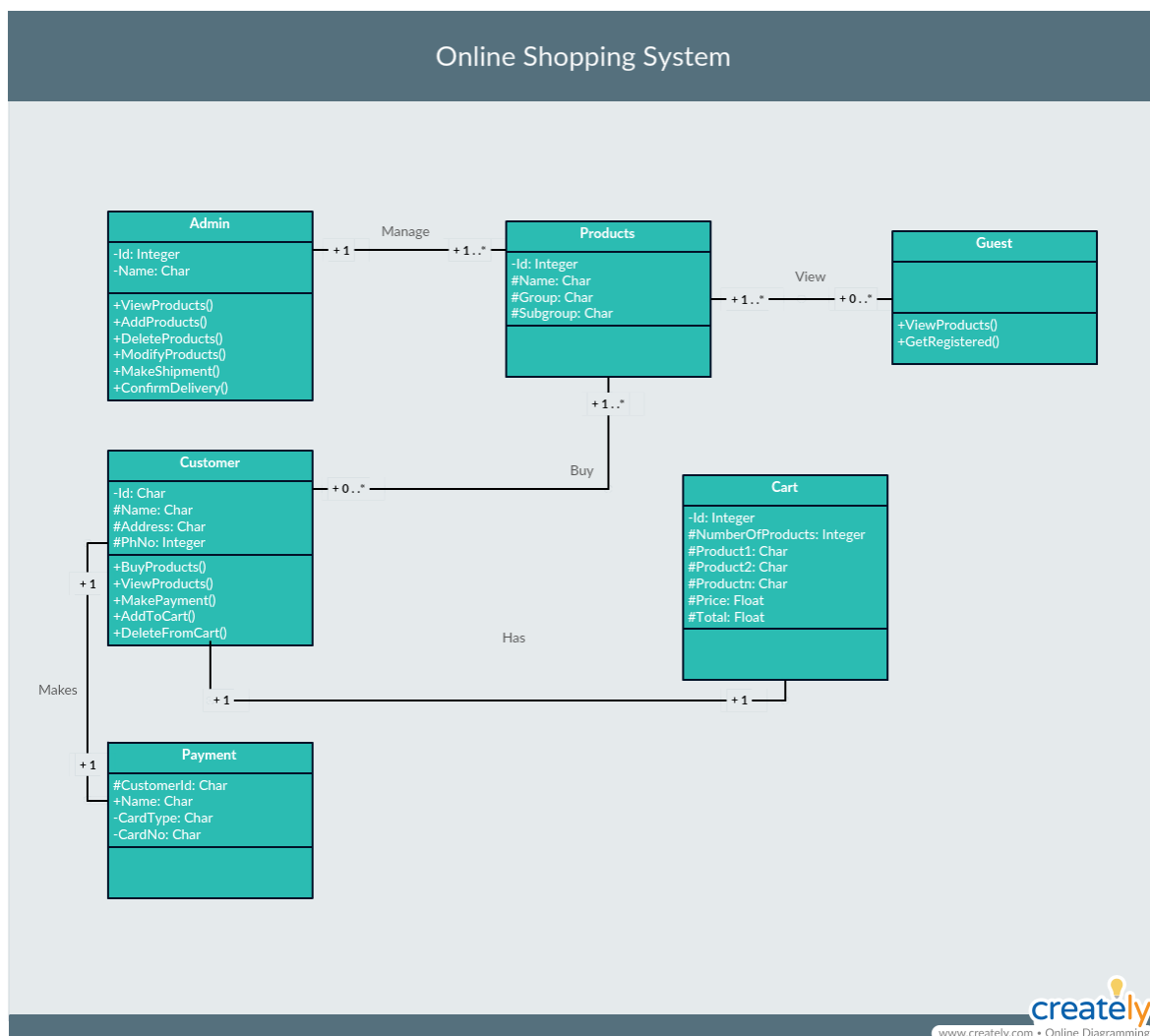
Class: Student

Attributes: rollNo, name

Methods: getDetails()

◆ Viva one-liner:

“Class Diagram represents the structure of the system and helps in object-oriented design.”



3 Sequence Diagram ★ ★ ★

◆ What is it?

A **Sequence Diagram** shows **step-by-step interaction** between objects in **time order**.

◆ Why do we use it?

- To understand **flow of messages**
- To show **how methods are called**

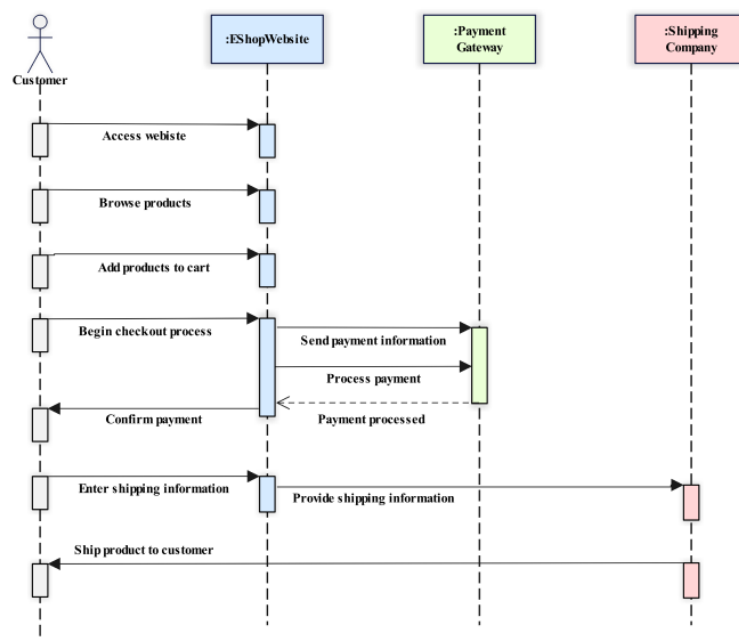
◆ Example:

User → Login → Authentication → Response

◆ Viva one-liner:

“Sequence Diagram is used to show interaction between objects in chronological order.”

Sequence Diagram for Online Shopping System



🔥 Activity Diagram ★★

◆ What is it?

An **Activity Diagram** shows the **flow of activities or actions**.

◆ Why do we use it?

- To represent **business logic**
- To show **workflow or process**

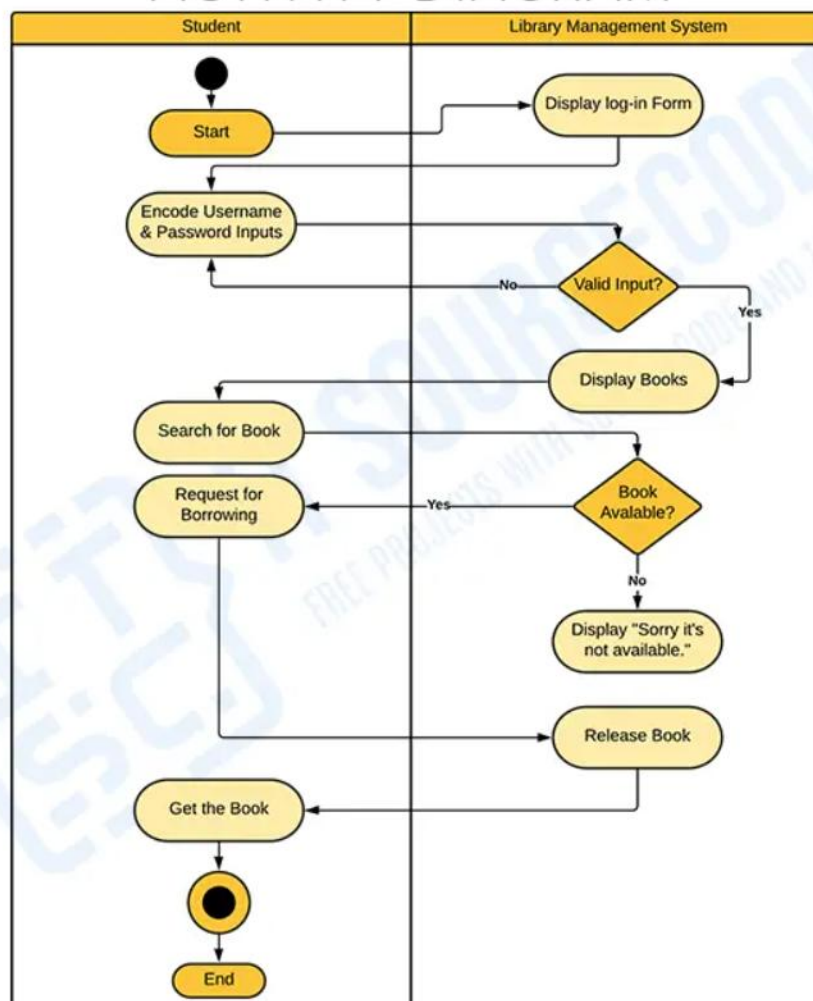
◆ Example:

Start → Login → Validate → Dashboard → End

◆ Viva one-liner:

“Activity Diagram represents the flow of control from one activity to another.”

LIBRARY MANAGEMENT SYSTEM ACTIVITY DIAGRAM



5 State Diagram ★★

◆ What is it?

A **State Diagram** shows **different states of an object** and transitions.

◆ Why do we use it?

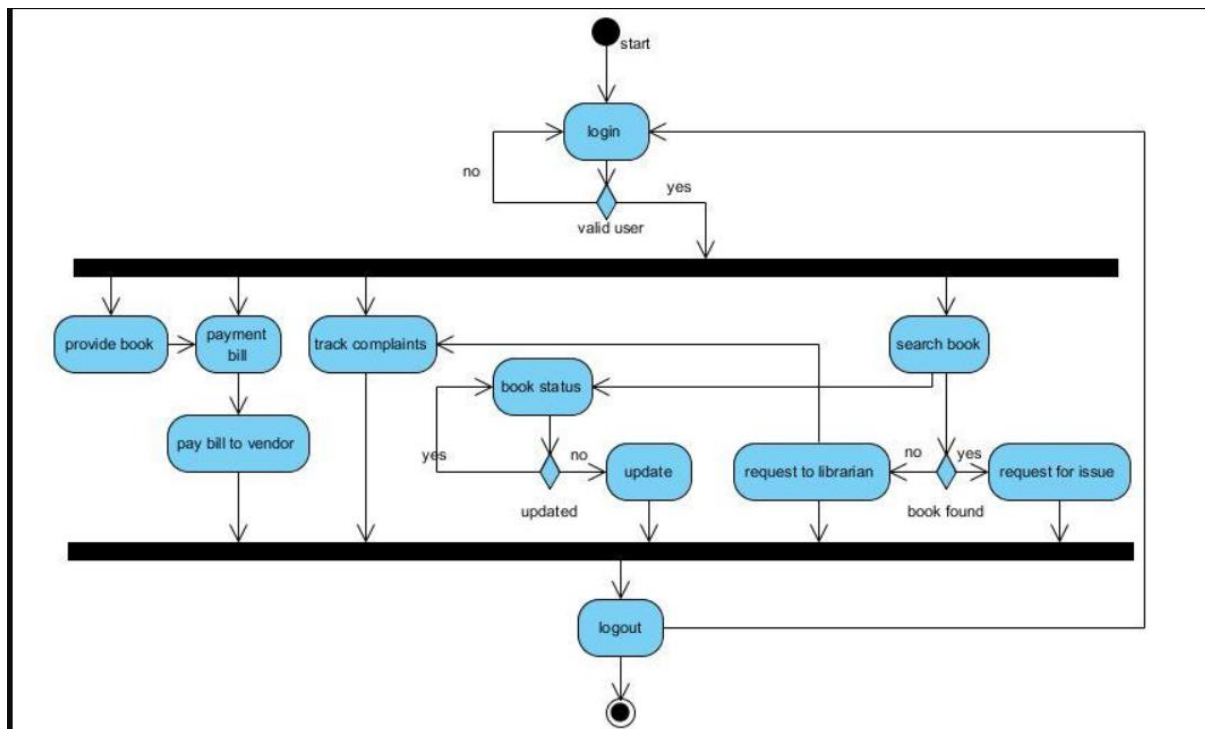
- To show **object behavior**
- Useful when object changes state

◆ Example:

Order: New → Confirmed → Shipped → Delivered

◆ Viva one-liner:

“State Diagram shows how an object changes its state based on events.”



6 Component Diagram ★ (Optional)

◆ What is it?

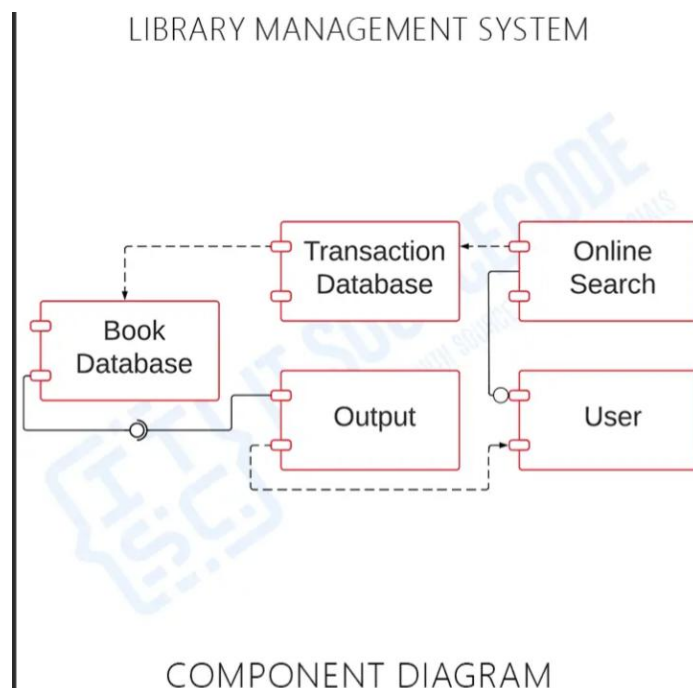
Shows **software components and their dependencies**.

◆ Why do we use it?

- To understand **system architecture**

◆ Viva one-liner:

“Component Diagram represents physical components of the system.”



7 Object Diagram

◆ What is it?

An **Object Diagram** shows **real objects (instances of classes)** at a particular time.

◆ Why do we use it?

- To show **actual data values**
- To understand **runtime behavior**

◆ Example:

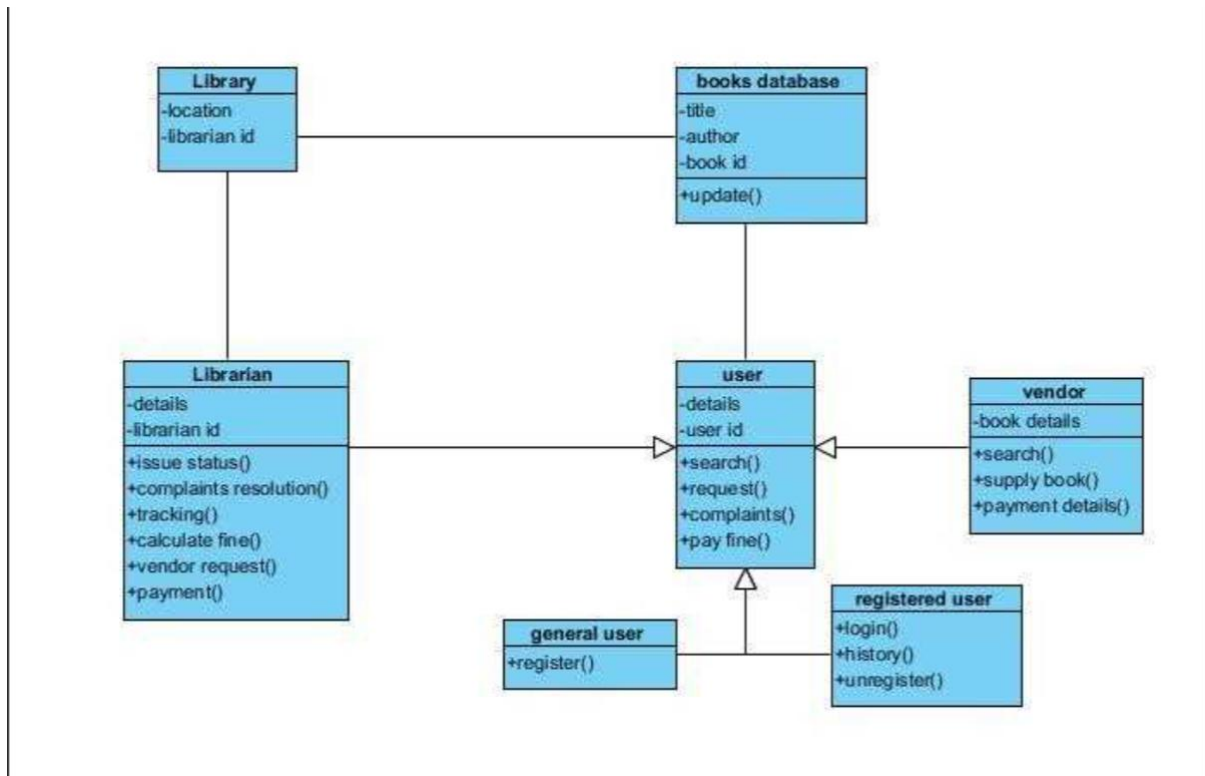
Student1

name = "Amit"

rollNo = 21

◆ **Viva one-liner:**

"Object Diagram represents instances of classes with actual values at runtime."



8 Communication Diagram (Collaboration Diagram)

◆ **What is it?**

A **Communication Diagram** shows **interaction between objects**, focusing on **message flow**.

◆ **Why do we use it?**

- To understand **object communication**
- Similar to sequence diagram but **structure-focused**

◆ **Example:**

User → Login → Auth → Database

◆ **Viva one-liner:**

"Communication Diagram shows how objects interact using messages."

9 Deployment Diagram

◆ What is it?

A **Deployment Diagram** shows **hardware and software deployment**.

◆ Why do we use it?

- To show **where application is deployed**
- Used for **system architecture**

◆ Example:

Client → Web Server → Database Server

◆ Viva one-liner:

“Deployment Diagram represents physical deployment of software on hardware nodes.”

10 Package Diagram

◆ What is it?

A **Package Diagram** groups **classes into packages**.

◆ Why do we use it?

- To organize large systems
- To show **dependencies between packages**

◆ Example:

package user
package admin

◆ Viva one-liner:

“Package Diagram is used to organize classes into logical groups.”

◇ ER (Entity Relationship) Diagram ★ ★ ★ (VERY IMPORTANT)

◆ What is it?

An **ER Diagram** represents **entities, attributes, and relationships** in a database.

◆ **Why do we use it?**

- To **design database structure**
 - To show **relationship between tables**
-

◆ **Main Components**

1 Entity → Real-world object

Example: Student, Course

2 Attribute → Properties of entity

Example: rollNo, name

3 Relationship → Connection between entities

Example: Student *enrolls* Course

◆ **Example:**

- Student (rollNo, name)
 - Course (courseId, courseName)
 - Relationship: Enrolls
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◆ **Cardinality (Viva favorite)**

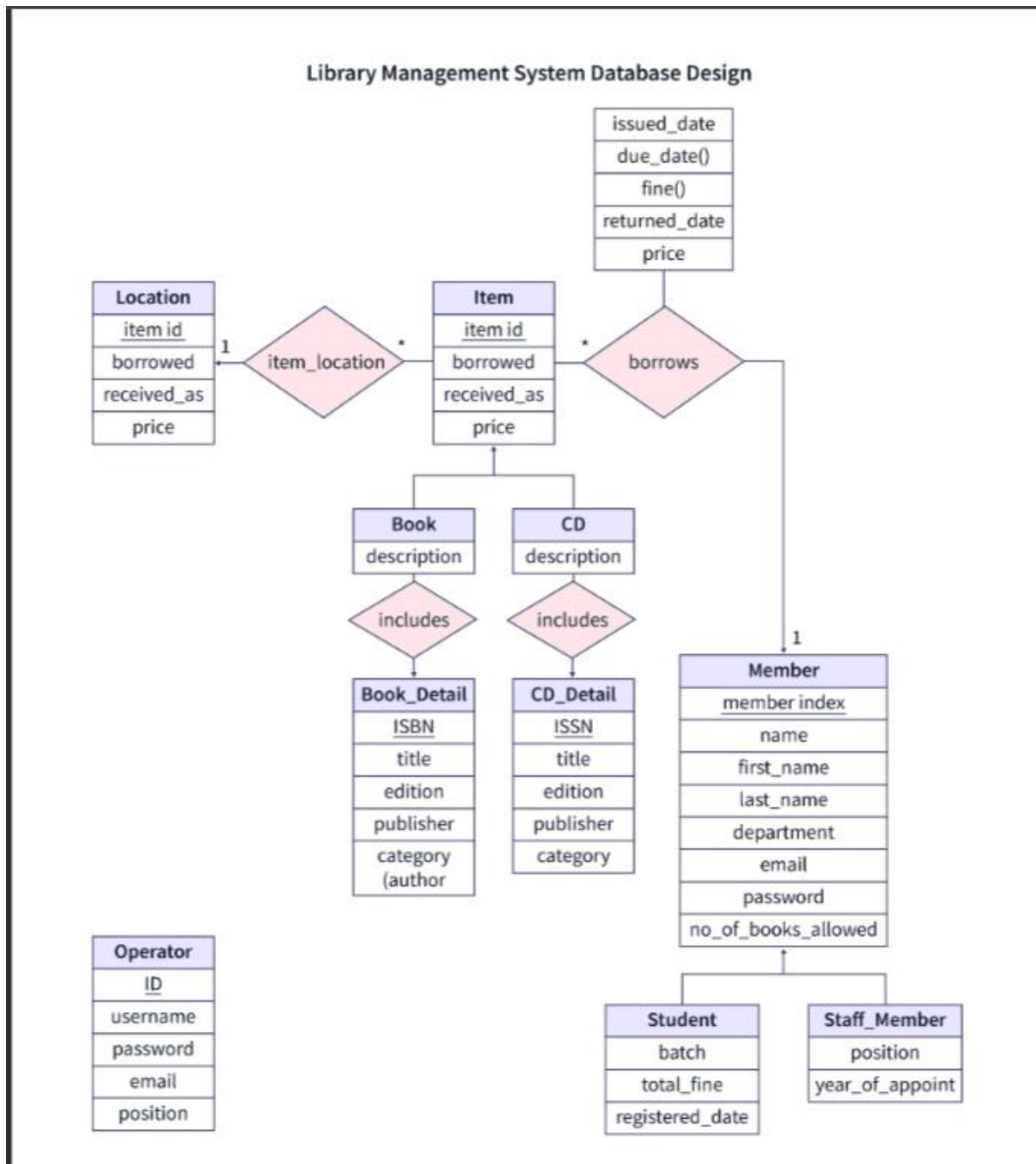
- **One-to-One (1:1)**
- **One-to-Many (1:M)**
- **Many-to-Many (M:N)**

Example:

One student can enroll in many courses → **1:M**

◆ **Viva one-liner:**

“ER Diagram is used to design database by showing entities, attributes, and their relationships.”



Diagram

Use Case

Class

Sequence

Activity

What it Shows

System functions

Classes & structure

Message order

Workflow

Why Used

Requirement analysis

Design & coding

Flow of interaction

Business logic

Diagram	What it Shows	Why Used
State	Object states	Behavior change
Object	Class instances	Runtime data
Communication	Object interaction	Message passing
Deployment	Hardware setup	Architecture
Package	Grouping	Code organization

★ ER Diagram vs Class Diagram (VERY COMMON QUESTION)

ER Diagram	Class Diagram
Database design	Software design
Entity, Attribute	Class, Method
Relationship	Association
No methods	Has methods